

Reading

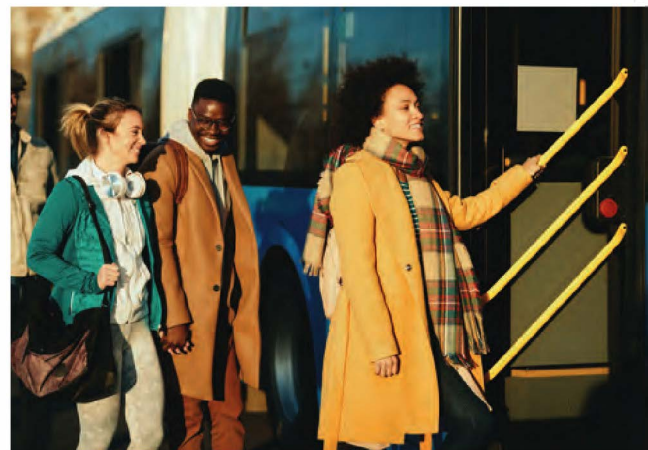
PREPARING TO READ

A BUILD VOCABULARY The words in **blue** are used in the reading passage. Read the text below. Then write the correct form of each word next to its definition.

Nobody likes a traffic jam. They lead to delays, **disruptions**, pollution, and stress. Unfortunately, congestion is a problem in most cities—the places with not just the most people, but the most people able to afford their own vehicles. Many cities don't have the **capacity** to handle all this traffic. Luckily, the solution is already at hand: People need to **swap** their cars for public transportation.

Advocates of public transportation cite several benefits. It is more efficient than cars, requiring much less fuel per passenger. It reduces the number of cars in a city, which lowers air, noise, and even light pollution levels. Furthermore, it lowers stress levels—fighting for parking spaces, for example, becomes a thing of the past. Plus, it reduces the cost of living in cities.

Clearly, though, most people still prefer driving over taking the bus or train. So what can we do to **urge** more people to embrace public transportation? The answer is simple: make it better. The sad reality is that public transportation is often underfunded and poorly planned. Vehicles are sometimes old and under-maintained. They don't run regularly enough, which results in **excessive** waiting times. And their routes often don't cover enough of the city to make them viable. It's important that public transportation officials work **alongside** city planners and invest sufficiently in solving these problems. It's worth noting that because public transportation is also a source of revenue, a good system ends up paying for itself.



1. _____ (adj) too much
2. _____ (n) the maximum something can contain or do
3. _____ (prep) together with
4. _____ (n) a person who publicly supports a cause or an idea
5. _____ (n) the disturbing or interruption of a process
6. _____ (v) to exchange one thing for another
7. _____ (v) to encourage someone to do something

B BUILD VOCABULARY Complete the sentences below with the correct form of the words in **blue**. Use a dictionary to help you.

aviation (n)

campaign (n)

catch on (phr)

committee (n)

competitive (adj)

conversion (n)

dilemma (n)

elite (adj)

1. Members of the _____ class in this city are unlikely to switch from cars to buses and trains, no matter how much public transportation improves.
2. Pilots and airport staff are just some of the people working in the _____ industry.
3. Car makers who sell only gas-powered vehicles will struggle to remain _____.
4. If prices continue to fall at this rate, it won't take long for EVs to _____ in this city.
5. They've assembled a(n) _____ to replan bus routes in the city.
6. The mayor launched a(n) _____ aimed at getting people to drive less.
7. The _____ of grain into biofuels such as ethanol is a relatively simple process.
8. Biofuels present us with a(n) _____: should we be turning food into fuel?

C USE VOCABULARY Note answers to the questions below. Then discuss with a partner.

1. Are there any **committees** you can join in your school or workplace? What are their goals?

2. How often do **disruptions** to public transportation occur in your town or city?

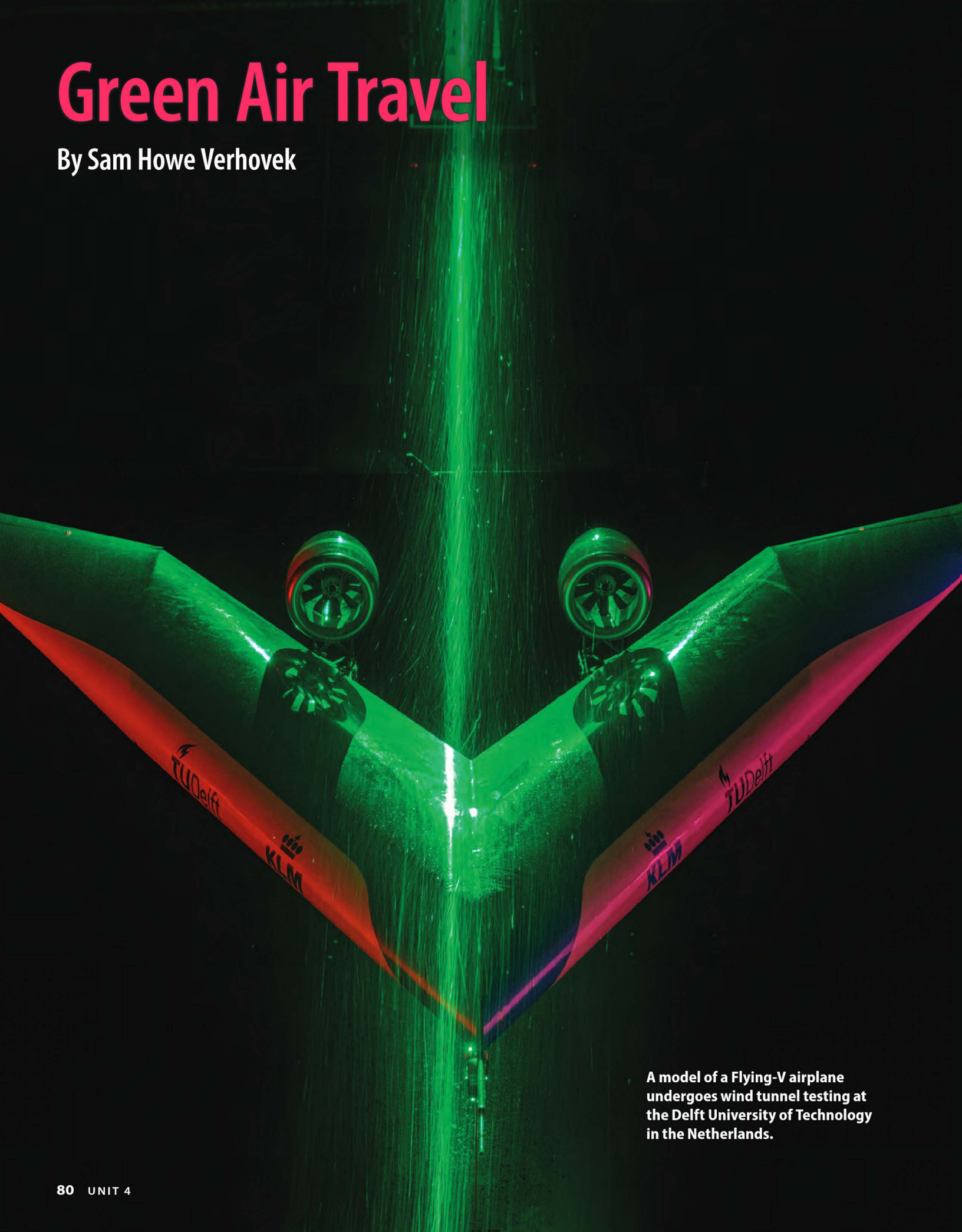
3. In a world that's becoming increasingly environmentally conscious, what are some things car makers could do to remain **competitive**?

D PREDICT Work with a partner. What are some things the aviation industry can do to make flying greener? Check your ideas as you read the passage.

Critical Thinking

Green Air Travel

By Sam Howe Verhovek



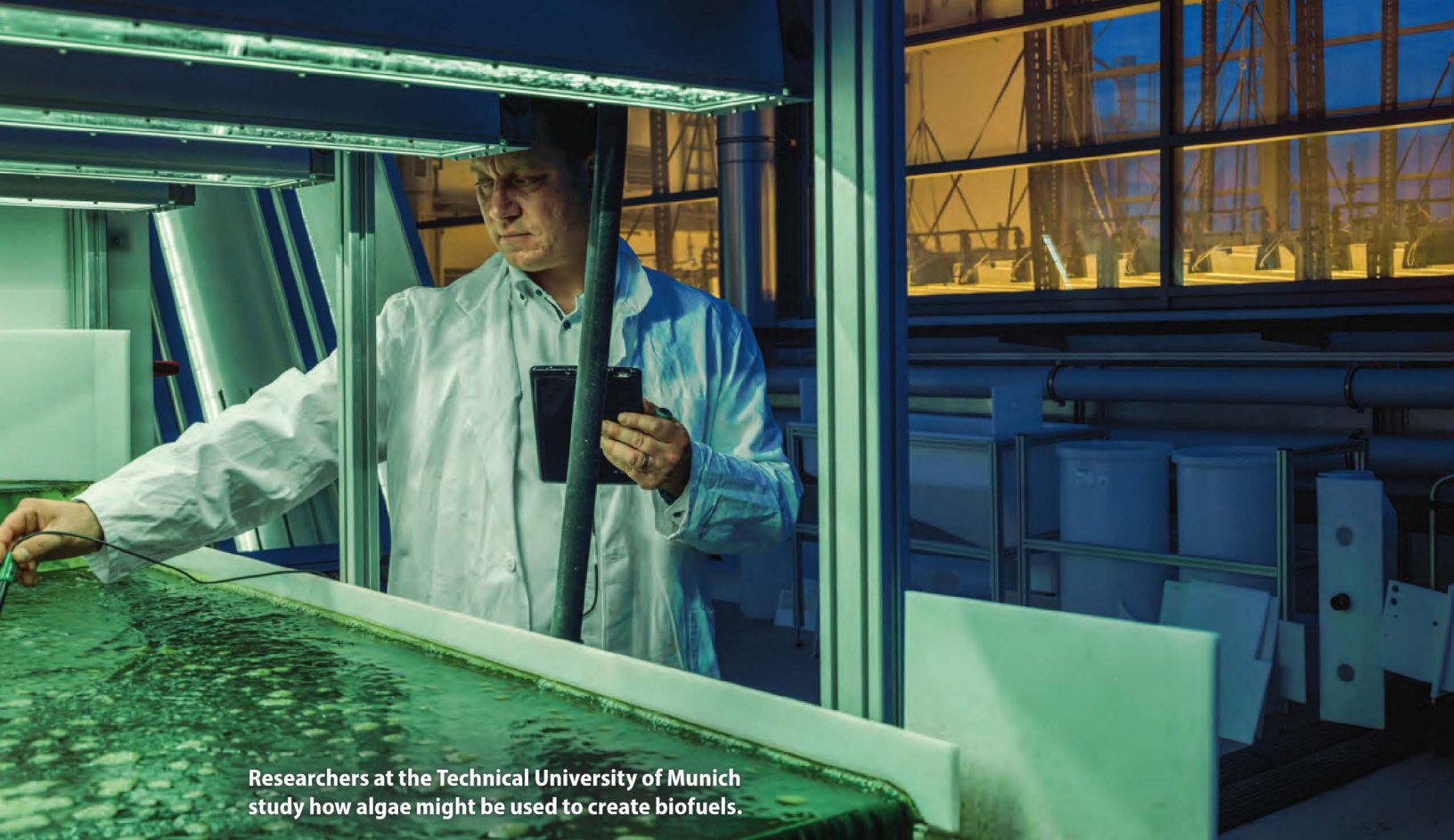
A model of a Flying-V airplane undergoes wind tunnel testing at the Delft University of Technology in the Netherlands.

Can the aviation industry reinvent itself to make flying less harmful to the environment?

- A**  As someone who loves to fly and never tires of looking at landmarks below, clouds **alongside**, or stars above, I can't begrudge anyone the joy of flight. At the same time, any journey in the skies warms the planet. Some experts peg air travel as the source of up to 5 percent of the human contribution to global warming today.
- B** That figure will likely climb as passenger and freight¹ air traffic grows, and as other activities like land transportation and construction become more energy efficient. All this has led to a movement **urging** people not to fly or at least to fly a lot less, a **campaign** with a name that has **caught on** in Europe and is becoming familiar elsewhere: *flygskam*, a Swedish term best translated as "flight shame."
- C** "Hour for hour, there is just about nothing you as an individual can do that's worse for the health of the planet than to sit on an airplane," says Peter Kalmus, an astrophysicist turned NASA climate scientist who hasn't flown since 2012. "The hard fact that most people haven't accepted yet is that we don't need to fly, and if you truly accept that we are in a climate emergency, you shouldn't fly."
- D** In July, France adopted a ban on all domestic air trips that can be made by train in less than two and a half hours. In the United Kingdom, the official **Committee** on Climate Change jolted the **elite** world of the most active fliers by proposing "a ban on air miles and frequent flyer loyalty schemes that incentivize **excessive** flying."
- E** However, **aviation** leaders contend that shaming flight is not the answer—greening it is.
- F** "Aviation is an essential part of the global economy, so our challenge is reducing emissions and decarbonizing aviation, not preventing people who want to travel from traveling," says Sean Newsum, the director of aviation sustainability strategy for Boeing. "That's really our foundational belief as an industry at this point."
- G** Among the potential paths to green salvation for air travel, the quickest might be down a gravel road deep in the woods of central Georgia, leading to a hulking complex called the Freedom Pines Biorefinery. There I meet Curt Studebaker, a lanky, friendly young chemical engineer who is in the business of turning waste—all kinds of waste—into sustainable aviation fuel (SAF).
- H** "The amazing thing is, once you get it right, it's really a better fuel even than Jet A," the standard kerosene fuel in U.S. aviation, Studebaker tells me. "It's actually cleaner."
- I** For now, SAFs are still blended with standard fuel. But they are cast as the giant first step toward shrinking aviation's carbon footprint.² The challenges? First, it's very expensive. This alternative fuel costs two to six times more than

¹Freight refers to cargo that is transported in bulk by land, air, or sea.

²Carbon footprint refers to the amount of greenhouse gases a person, group, or industry releases into the atmosphere.



Researchers at the Technical University of Munich study how algae might be used to create biofuels.

kerosene, and although more flights are using SAFs, it all adds up to little more than a drop in the bucket—well under 0.1 percent of the 95 billion gallons of fuel the industry used in 2019. Second, the industry can't rely on the easiest, cheapest sources for **conversion**: crops. If fuel producers were to gobble up land and water more urgently needed for food, air travel would simply trade one environmental black eye for another.

J Proponents contend that if SAF production were built to the scale needed to serve the bulk of aviation needs, the price would drop precipitously,³ becoming **competitive** with kerosene. But getting to scale is a classic chicken-or-egg **dilemma**. Unless there's demand, supply won't grow; but because the current supply is so small and costly, it's hard to stimulate demand. That's where the problem becomes political: the solution could be a carbon tax on kerosene or a requirement that SAFs account for a percentage of all aviation fuel.

K “Basically, there has to be a humongous ramp-up to SAFs,” says Paul Stein, chief technology officer of Rolls-Royce, the British manufacturer whose next-generation UltraFan, the biggest and one of the most efficient jet engines ever, is designed to use the alternative fuel. “But industry is generally behind a SAF mandate.⁴ And certainly our position as a company is, yes! We need more SAFs. It would be a huge contribution to the planet.”

³ If something drops **precipitously**, it drops quickly and by a large amount.

⁴ A **mandate** is an official ruling or order passed on from a high authority.

L At Airbus headquarters in the south of France, there is a flying machine made of composite materials⁵ resembling no airliner that has ever taken to the skies, at least outside of science fiction movies or UFO⁶ sightings. The plane, known as Maveric, is a model aircraft with a 3.2-meter wingspan. For Airbus, Maveric's design could hold the answer to this intriguing question: Is there a more efficient—greener—way to design an airliner?

M For all kinds of reasons, the modern aircraft manufacturing industry does not easily lend itself to the **disruption** that can so suddenly upend conventional thinking in other industries. A true game changer of an airliner will take many, many years to develop and more years to weather the gauntlet of safety tests involved in certification for commercial service.

N Yet the so-called blended wing body design employed by Maveric—although with major technical challenges to overcome—could yield as much as a 40 percent reduction in carbon emissions compared with today's planes. The main advantage of the streamlined design is that the entire aircraft functions much like a wing, reducing drag⁷ and making it much easier to generate lift. In the Netherlands, researchers at the Delft University of Technology used similar principles in designing Flying-V, an aircraft that looks very much like a boomerang.⁸

⁵ Composite materials are made by combining materials with different properties.

⁶ UFO stands for "unidentified flying object." The term refers to aircraft that are unrecognizable or thought to be from outer space.

⁷ Drag is the wind resistance that airplanes and other vehicles experience when moving at speed.

⁸ Boomerangs are small V-shaped objects from Australia that return when thrown correctly.



A model of Airbus's Maveric, with its blended wing design

- o In 2020, Airbus went a step further and created a major stir in the industry by announcing it was working on a line of aircraft with a stunning capability: zero-emission flight. A Maveric variant and two smaller tube-and-wing airliners, it said, would operate on hydrogen fuel. The main by-product? Water vapor.
- P As is true with electric automobiles, zero emission doesn't necessarily mean zero pollution. Just as it matters where the electricity comes from to charge the car's battery, Airbus's approach begs the question of how to create and store hydrogen fuel.
- Q Most hydrogen used today comes from fossil fuels. But so-called green hydrogen, in which clean electricity is used to separate water into hydrogen and oxygen, is the holy grail. **Advocates** say that technological progress and scaling up will bring green hydrogen its day in the sky.
- R In the central California farm town of Hollister, a stubby banana-yellow aerial vehicle with 13 rotors whirls around. It has no pilot.
- S The self-flying electric plane may be an oddity today, but its inventors expect it to be a commonplace feature of tomorrow—the aerial taxi. As more than one evangelist for the urban air mobility industry puts it, “Think: Uber meets Tesla in the sky.”
- T Their company, called Wisk, is just one of many aspiring entrants, although with major chops: It has financial backing from Boeing and Kitty Hawk, the aviation start-up founded by Google's Larry Page. Its vision is a world in which taking a flying taxi will be as easy and affordable as an automobile ride is today—and safer to boot.
- U “This is not the Wild West,” Gary Gysin, Wisk's chief executive, tells me when I visit the company's hangar.⁹ “We will absolutely meet the incredibly stringent safety standards already set for the aviation industry. We have to—nobody's flying anywhere until the FAA (the Federal Aviation Administration) says so.”
- V Just when this particular industry might take off is, well, up in the air. Gysin says the industry likely will start by shuttling people among airports and “vertiports,” which might be a landing pad atop a Manhattan apartment building, or a parking lot in a Los Angeles suburb.
- W Just how strong a public backlash to the idea of air taxis there might be is hard to say. But electric-powered flight, while still severely limited by battery weight and **capacity**, is happening on another front. One intriguing approach is in British Columbia, Canada, where a commuter seaplane operator is retrofitting¹⁰ its workhorse fleet of 60-year-old de Havilland Beavers and Otters, **swapping** out gas-fired piston engines for electric motors.

⁹ A **hangar** is a place where airplanes are kept.

¹⁰ **Retrofitting** is the act of customizing a finished product by installing new parts, in a way not intended by the manufacturer.

Harbour Air's battery-powered sea plane gets its power from a hydro-electric dam.



- x Greg McDougall, Harbour Air's founder and chief executive, piloted the December 2019 initial test run on the first such plane. "We're proud to be the first airline in the world to offer completely clean electric flight, fueled by our province's sustainable hydropower," McDougall tells me. "But I'm not doing this just because I'm some wild-eyed environmentalist hippie. I am a businessman. This is going to lower my costs, which is going to lower the cost of everyone's tickets."

Adapted from "How Green, How Soon?" by Sam Howe Verhovek: National Geographic Magazine, October 2021

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UNDERSTANDING THE READING

A UNDERSTAND MAIN IDEAS Choose the correct heading for each section in the reading passage.

- | | |
|------------------------|-----------------------------------|
| 1. ____ Paragraphs A–F | a. Electric Flights |
| 2. ____ Paragraphs G–K | b. Using Greener Energy |
| 3. ____ Paragraphs L–O | c. Air Travel and the Environment |
| 4. ____ Paragraphs R–X | d. Designing Better Planes |

B UNDERSTAND DETAILS Complete the sentences. Use no more than two words from the reading passage for each answer.

- Freedom Pines Biorefinery uses _____ to create sustainable aviation fuel.
- Rolls-Royce produced a large and _____ jet engine that is designed to run on alternative fuel.
- Airbus is designing airplanes that experience less _____ and more _____.
- Wisk wants its planes to function as _____ that fly.
- The FAA ensures that airplanes meet _____.
- Harbour Air uses clean electricity that is generated using _____.

C UNDERSTAND DETAILS Note answers to the questions below. Then discuss with a partner.

- What is the strategy of the *flygskam* campaign?

- Why do you think some countries have proposed a ban on frequent flyer loyalty schemes?

- How can government intervention make SAFs more viable?

- What has to happen for hydrogen to be considered green?

- Why might there be a public backlash against air taxis?

